

ABHISHEK S

abhishek.shapelly3@gmail.com | +19034599999 | [LinkedIn](#)

PROFESSIONAL SUMMARY

Senior Data Engineer with 8+ years of experience designing and delivering enterprise data engineering solutions within the Microsoft Azure ecosystem, including **Azure Data Factory, Azure Databricks, ADLS**, and Synapse Analytics. Expertise in **SQL, Python, and Apache Spark, building scalable ETL pipelines** and cloud-native data platforms to **support analytics, reporting**, and business intelligence.

Proven experience architecting **data lakehouse solutions, implementing dimensional data models** (star schema), and building metadata-driven data pipelines for large-scale enterprise environments. Strong background in integrating **API-based data sources, handling JSON transformations, and ensuring data quality**, governance, and security across data platforms.

Experienced in working in client-facing **consulting environments, translating business requirements into scalable data architectures**, and delivering best-in-class data solutions aligned with business goals. Adept at mentoring teams, leading design sessions, and driving data engineering strategy, automation, and performance optimization in Agile environments.

TECHNICAL SKILLS:

Hadoop/Big Data Technologies: HDFS, Apache NiFi, MapReduce, Sqoop, Flume, Pig, Hive, Oozie, Impala, Zookeeper, Ambari, Storm, Spark, Kafka

NoSQL Databases: HBase, Cassandra, MongoDB

Monitoring & Reporting: Tableau, Custom Shell Scripts, Power BI

Hadoop Distributions: Hortonworks, Cloudera, MapR

Build & Deployment Tools: Maven, SBT, Git, SVN, Jenkins, Kubernetes

Programming & Scripting: Scala, Java, SQL, Python, JavaScript, Shell Scripting, Pig Latin, HiveQL

Databases: Oracle, MySQL, MS SQL Server, Vertica, Teradata, Azure Data Warehouse

Analytics Tools: Tableau, Microsoft SSIS, SSAS, SSRS

Web Development Technologies: HTML, XML, JSON, CSS, jQuery, JavaScript

IDE & Development Tools: Eclipse 3.5, NetBeans, MyEclipse, Oracle JDeveloper 10.1.3, Ant, Maven, RAD

Operating Systems: Linux, Unix, Windows 7/8, Windows Server 2003/2008

AWS Services: EC2, EMR, S3, Redshift, Lambda, Glue, Data Pipeline, Athena, Elasticsearch, SQS, DynamoDB, Kinesis, RDS, EBS

Azure Cloud Platforms & Services: Databricks, Data Lake, Blob Storage, Azure Data Factory, SQL Database, SQL Data Warehouse, Cosmos DB, Active Directory

Network Protocols: TCP/IP, UDP, HTTP, DNS, DHCP

PROFESSIONAL EXPERIENCE

Ever north(Cigna) — Senior Data Engineer

Feb 2024 –Present

- Designed and implemented **scalable ETL pipelines using PySpark, SQL, Azure Databricks, and AWS**, enabling **enterprise data integration and analytics at scale**
- Architected **data lakehouse solutions and dimensional data models (star schema)** to support **reporting, BI, and analytics use cases**
- Developed **metadata-driven data pipelines and reusable frameworks**, improving **pipeline maintainability and scalability across projects**
- Integrated **API-based data sources and processed JSON data**, transforming and normalizing into structured formats for downstream consumption
- Implemented **data quality frameworks, validation checks, and governance controls**, ensuring **data reliability and consistency**
- Collaborated with stakeholders and business teams to gather requirements and deliver **data solutions aligned with business objectives**

UnitedHealth Group (Optum) — Senior Data Engineer

Sep 2023 – Jan 2024

- Engineered high-performance Spark applications (Python, PySpark, Spark SQL) to process large-scale data and load into Hive and Azure SQL, enabling reliable downstream analytics.
- Architected scalable ETL/data pipelines using Python, PySpark, Azure Databricks, and Azure Data Factory (ADF) to support large-scale distributed data processing across enterprise datasets.
- Automated end-to-end ETL workflows and environment setup using Python, PySpark, and Terraform (IaC) to provision and manage Azure Data Lake, Storage Accounts, and Databricks clusters, ensuring consistent, repeatable deployments.
- Established CI/CD practices with GitLab, implementing version control, code reviews, and automated build/test/deploy pipelines for ADF and Databricks jobs, improving release quality and deployment speed.
- Optimized Spark performance using partitioning strategies, broadcast joins, and caching, efficiently processing multi-terabyte datasets while reducing runtime and compute costs.
- Developed data quality frameworks and Snowflake data models, writing optimized SnowSQL for transformations, aggregations, and event-based triggers, achieving ~40% query performance improvement and delivering trusted analytical datasets in collaboration with business stakeholders and data scientists.

Apollo Global Management — Senior Data Engineer

Apr 2023 – Aug 2023

- Architected end-to-end ETL pipelines using Azure Data Factory (ADF) and Terraform, provisioning and managing Azure Data Lake, Blob Storage, and Databricks to support scalable, secure data platforms.
- CI/CD deployments with GitLab for ADF and Databricks, reducing manual release effort by 50% and improving deployment consistency and speed.
- Optimized PySpark workloads on Azure Databricks, delivering a 40% reduction in execution time and compute costs through performance tuning and distributed processing best practices.
- Designed high-performance Snowflake data models and optimized SQL queries, achieving a 25% improvement in BI query efficiency for analytics and reporting teams.
- Integrated multi-source data pipelines (SQL databases, cloud storage, REST APIs) using Python and PySpark, with monitoring, logging, and alerting frameworks that ensured reliability and proactive issue resolution via ServiceNow and JIRA.

Capital One — Senior Data Engineer

May 2022 – Apr 2023

- **Big Data & ETL Development:** Built scalable ETL pipelines using Spark (Scala/PySpark), Databricks, and AWS EMR for data transformation, aggregation, and cleansing. Automated data onboarding, tokenization, encryption, and delivery to AWS S3.
- **Cloud & Data Platform Expertise:** Leveraged AWS services (S3, Lambda, Glue, Redshift, KMS, EC2) for secure data processing and storage. Integrated Snowflake for role-based access, secure sharing, and cost optimization.
- **Workflow Orchestration & Automation:** Developed Airflow DAGs for batch processing and Spark job scheduling. Built CI/CD pipelines with Jenkins, Docker, Kubernetes, ensuring secure microservices deployments.
- **Data Analytics & Reporting:** Performed analytics on large datasets using PySpark, Scala, and Spark-SQL, generating partner reports. Handled bad, null, and partial records to ensure high-quality data.
- **Performance Optimization & Scalability:** Implemented Delta Lake for efficient concurrent operations and rollbacks. Optimized Redshift queries and orchestrated Kubernetes infrastructure, reducing processing time by 30%.

Wells Fargo — Senior Data Engineer

Aug 2018 – Sep 2019

- Designed and implemented Hadoop-based Big Data solutions and custom applications on Azure platforms, including Data Factory, Data Lake, SQL Database, SQL DW, HDInsight, and Databricks. Built ADF pipelines to ingest and transform enterprise data from on-premises to Azure Data Lake, Blob Storage, and Synapse Analytics.

- Developed **complex ETL pipelines** and **batch/real-time data ingestion** using **PySpark, Spark-SQL, Hive, MapReduce, Sqoop, and shell scripting**. Performed **data extraction, transformation, aggregation, cleansing**, and loaded massive datasets into **HDFS, HBase, and Delta Lake**.
- Executed **ad-hoc analysis** using **Azure Data Lake Analytics, HDInsight, Databricks, Spark, and Zeppelin/Jupyter notebooks**. Created **tabular models** on **Azure Analysis Services**, SSIS modules, and BI solutions using **SSIS, SSRS, SSAS** for business reporting.
- Created **Oozie workflows, ADF-Databricks integration using widgets**, and automated **Spark job scheduling**. Managed and provisioned **Databricks clusters** and **HDInsight Spark clusters**, including library installations, cluster sizing, monitoring, and troubleshooting.
- Worked with **Delta Lake, Spark Streaming, Kafka, and Kinesis** for real-time data processing. Implemented **Hive tuning, partitioning, bucketing, and memory optimization**, enabling efficient processing of **semi-structured and large-scale datasets**, ensuring high-quality, scalable data solutions.

Ness Technologies — Data Engineer

Sep 2017 – Aug 2018

- **Architected and implemented Hadoop pipelines** using **MapReduce, PySpark, Hive, Pig, and Oozie**, including **backup, recovery, data ingestion, and automation**. Migrated legacy MapReduce jobs to **PySpark** for faster processing.
- **Designed and optimized data storage and processing** in **Hive, Teradata, and AWS Redshift**, handling **table creation, indexing, statistics, and data migration** with **Sqoop, JDBC, FastLoad, and Spark SQL**.
- **Developed scalable data solutions** with **Scala/Java UDFs, Spark DataFrames, Kafka, and Solr/Lucene**, working on **XML, JSON, Avro, Sequence, and Map files** for structured data processing.
- **Automated workflows and managed production systems** using **Oozie, Zookeeper, UNIX scripts, and AWS (EC2, S3, EMR)**, including **code deployment, issue resolution, log monitoring, and pipeline orchestration**.
- **Performed advanced data analysis and reporting** using **AWS Redshift, Power BI, and Spark SQL** to generate insights on **sales, products, and marketing**, collaborating with **support teams** for troubleshooting and optimizations.

Infosys — Software Engineer

Sep 2015 – Sep 2017

- **Designed, developed, and validated solutions** by analyzing client requirements, creating detailed designs, and collaborating with client teams; involved in **data extraction and export from DB2 to AWS** for analysis, reporting, and visualization.
- **Built and optimized big data pipelines** using **Spark, Scala, Kafka, Spark Streaming, HDFS, Hive, and Impala**; performed **complex ETL, data transformations, and migrations (AWS → MySQL)**, and converted **SQL queries to Spark RDD transformations**.
- **Implemented real-time data workflows and REST APIs**, including **Oozie job orchestration, HBase table creation, log collection from web servers, and continuous ETL pipelines**, ensuring efficient **data management and analysis**.

EDUCATION

MS | Business Analytics | University of North Texas | GPA: 3.6

B.Tech | Chemical Engineering | Osmania University | GPA: 3.5